ELG 2186

Lecture 2

Analyzing circuits with diodes

at most 3 diodes per circuit in this course.

1 diode -> 2 possibilities → ON / → OFF

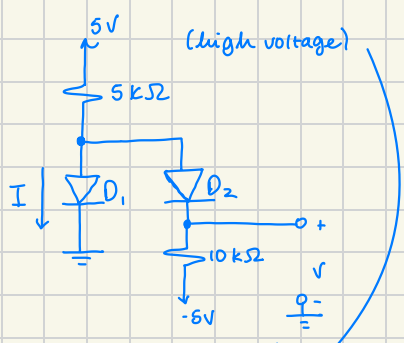
2 diodes, say D_1 & D_2 -> 4 possibilities ->

D_1	D_2
OFF	OFF
OFF	ON
ON	OFF
ON	ON

3 diodes, say D_1, D_2, D_3 -> 6 possibilities ->

D_1	D_2	D_3
OFF	OFF	OFF
OFF	OFF	ON
OFF	ON	OFF
OFF	ON	ON
ON	OFF	OFF
ON	OFF	ON
ON	ON	OFF
ON	ON	ON

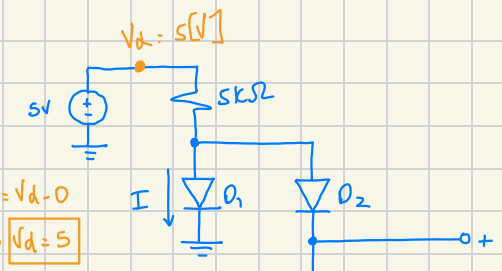
Example: Calculate the value of the current I if the diodes are ideal



current flows from high to low voltage.

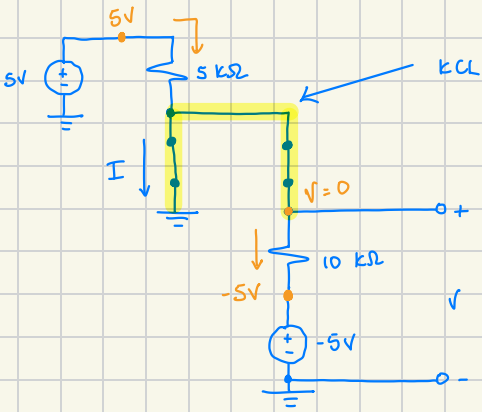
Hence assume, D_1: ON, D_2: ON since diodes are ideal: D_1 & D_2 are short-circuits.

Circuit can also be drawn as:



$5 = V_d - 0$ -> $\boxed{V_d = 5}$

$-5 = V_\beta - 0$ => $\boxed{V_\beta = -5V}$



KCL : $\frac{5-0}{5000} = I + \frac{0-(-5)}{10\,000}$

$\Rightarrow \frac{5}{5000} = I + \frac{5}{10\,000}$

$\Rightarrow I = \frac{5}{5000} - \frac{5}{10\,000} = \frac{1}{2000} = 0.0005\,[A]$

$\Rightarrow \boxed{I = 0.0005[A] = 0.5 \times 10^{-3}\,[A] = 0.5[mA]}$